

Homework 9

Chapter 19

1)

```
#ifndef QUEUE_H
```

```
#define QUEUE_H
```

```
typedef struct Queue {
```

```
    int *items;
```

```
    int front;
```

```
    int rear;
```

```
    int capacity;
```

```
    int size;
```

```
} Queue;
```

```
void initQueue(Queue *q, int capacity);
```

```
void enqueue(Queue *q, int item);
```

```
int dequeue(Queue *q);
```

```
int front(Queue *q);
```

```
int rear(Queue *q);
```

```
int isEmpty(Queue *q);
```

```
int isFull(Queue *q);
```

```
void freeQueue(Queue *q);
```

```
#endif // QUEUE_H
```

Chapter 20

4)

```
#define MK_COLOR(red, green, blue) ((long)((red & 0xFF) << 16) | ((green & 0xFF) << 8) |  
(blue & 0xFF))
```

6.A)

```
#include <stdio.h>
```

```
unsigned short swap_bytes(unsigned short i) {  
    return (i >> 8) | (i << 8);  
}
```

```
int main() {  
    unsigned short num;  
  
    // Read a hexadecimal number (up to four digits)  
    printf("Enter a hexadecimal number (up to four digits): ");  
    scanf("%hx", &num);  
  
    // Swap bytes and print the result  
    unsigned short swapped = swap_bytes(num);  
    printf("Number with bytes swapped: %hx\n", swapped);  
  
    return 0;  
}
```

6.B)

```
unsigned short swap_bytes(unsigned short i) {  
    return (i >> 8) | (i << 8);  
}
```

7)

```
#include <stdio.h>
```

```
unsigned int rotate_left(unsigned int i, int n) {  
    return (i << n) | (i >> (32 - n));  
}
```

```
unsigned int rotate_right(unsigned int i, int n) {  
    return (i >> n) | (i << (32 - n));  
}
```

```
int main() {  
    unsigned int num;  
    int n;  
  
    printf("Enter an unsigned integer (in hexadecimal): ");  
    scanf("%x", &num);  
  
    printf("Enter the number of positions to rotate: ");  
    scanf("%d", &n);  
  
    unsigned int left_rotated = rotate_left(num, n);
```

```
unsigned int right_rotated = rotate_right(num, n);

printf("Left rotated: 0x%x\n", left_rotated);
printf("Right rotated: 0x%x\n", right_rotated);

return 0;
}
```

Chapter 21

7.A)

A function that determines the current day of the week

- Expected Header: <time.h>
- The function localtime() or gmtime() could be used with additional code to extract the current day of the week. While there is no standard function that directly gives the day of the week, these time functions allow you to obtain and manipulate time data.

7.B)

A function that tests whether a character is a digit

- Expected Header: <ctype.h>
- The function isdigit() checks whether a character is a digit.

7.C)

A macro that gives the largest unsigned int value

- Expected Header: <limits.h>
- The macro UINT_MAX gives the maximum value for an unsigned int.

7.D)

A function that rounds a floating-point number to the next higher integer

- Expected Header: <math.h>
- The function `ceil()` rounds a floating-point number up to the next integer.

7.E)

A macro that specifies the number of bits in a character

- Expected Header: <limits.h>
- The macro `CHAR_BIT` specifies the number of bits in a char.

7.F)

A macro that specifies the number of significant digits in a double value

- Expected Header: <float.h>
- The macro `DBL_DIG` specifies the number of significant digits for a double.

7.G)

A function that searches a string for a particular character

- Expected Header: <string.h>
- The function `strchr()` searches for the first occurrence of a character in a string.

7.H)

A function that opens a file for reading

- Expected Header: <stdio.h>
- The function `fopen()` opens a file for reading (using the "r" mode).